

Tyler Davis

Senior Data Engineer

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Summary

- Around 5 years of Data Engineering experience with expertise in designing, building, and optimizing data pipelines and architectures for large-scale data processing.
- Skilled in Big Data technologies such as Hadoop, HDFS, Apache Spark, and Kafka, with hands-on experience in data processing frameworks including Spark SQL and Spark Streaming.
- Strong background in programming languages such as Python, Java, and Scala, as well as SQL and shell scripting for automating data workflows.
- Proficient in data pipeline automation using CI/CD and orchestration tools such as Jenkins, Apache Airflow, and Docker, ensuring reliable and efficient data workflows.
- Skilled in data visualization and analytics using Power BI and Tableau, as well as Python libraries like Matplotlib and Seaborn, to generate insights and support data-driven decision-making.
- Demonstrated strong analytical and diagnostic skills with experience in gathering comprehensive project requirements and applying SDLC models, including Agile, Waterfall

Skills

- **Big Data Technologies:** Hadoop, MapReduce, HDFS, Apache Spark, Kafka, Hive, HBase
- **Data Processing and Streaming:** Spark SQL, Spark Streaming, Kafka, MapReduce
- **ETL Tools:** Informatica PowerCenter, Apache NiFi, SQL Server Integration Services (SSIS)
- **Languages:** Python, Java, Scala, SQL, Shell Scripting
- **Azure Services:** Azure Data Factory, Azure Blob Storage, Azure ML Studio, Azure Functions, Azure HDInsight, Azure SQL Database
- **AWS Services:** AWS EMR, EC2, S3, RDS, Redshift, Data Pipeline, Lambda
- **GCP Services:** Dataproc, BigQuery
- **CI/CD Tools:** Jenkins, Git, Azure DevOps, Docker
- **Version Control:** Git, GitHub, Bitbucket
- **Configuration & Orchestration:** Docker, Terraform, Ansible, Airflow
- **Relational Databases:** MS SQL Server, MySQL, PostgreSQL, Teradata
- **NoSQL Databases:** HBase, Cassandra, MongoDB
- **Cloud Databases:** Amazon Redshift, Azure SQL, Database, Azure Synapse
- **Machine Learning Libraries:** Spark MLlib, ScikitLearn, TensorFlow
- **Visualization:** Power BI, Tableau, Matplotlib, Seaborn
- **Monitoring Tools:** Splunk, Elasticsearch, Grafana

Experience

Senior Data Engineer | July 2024 - Present

Capgemini | New York, NY

- Optimized Kafka-based architecture to handle over 1 million records per day, enabling efficient real-time data processing and downstream consumption by Apache Spark for transformation and analytics.
- Optimized policy data transformation using Apache Spark, implementing efficient formatting, filtering, and aggregation techniques, reducing transformation time by 30% and accelerating processing.

- Migrated policy and claims data processing from traditional SQL databases to Delta Lake on Databricks, boosting data processing speed by 50% and reducing infrastructure costs by 40% through optimized storage and batch processing.
- Developed scalable batch and stream processing workflows in Databricks, integrating Kafka streams and historical data into Delta tables for real-time insights and enhanced decision-making.
- Automated ETL workflows using Azure Data Factory, reducing manual intervention by 25% and ensuring consistent, timely data updates from 10+ data sources.
- Optimized SQL Server and PostgreSQL databases for improved query performance, reducing retrieval time by 20% using indexing and partitioning strategies.
- Developed Python-based automation scripts to validate data integrity during transformation and loading, reducing errors by 10% and ensuring accurate data.
- Deployed and automated end-to-end data pipelines with Jenkins, Docker for containerization, and Terraform for cloud infrastructure management, reducing deployment times by 40%.
- Developed interactive Power BI dashboards to visualize real-time policy and claims data, enabling business stakeholders to track key metrics and make data-driven decisions.

Data Engineer | Jan 2018 - Jun 2024

American International Group | New York, NY

- Optimized batch data processing pipelines with Apache Spark and HDFS, improving processing speeds by 30% through efficient configurations, repartitioning, and caching intermediate results.
- Streamlined real-time financial data ingestion with Apache Kafka, optimizing data formats like Avro for faster processing and reducing network latency when handling millions of transaction records daily.
- Enhanced ETL workflows in Informatica PowerCenter, introducing incremental loading to reduce ETL run times and ensure efficient data transformations for daily banking reports.
- Designed data archiving strategies using S3 Glacier and HDFS tiering, cutting storage costs by 25%.
- Optimized AWS Lambda functions for serverless processing of financial transaction data, improving scalability during peak periods and reducing operational costs.
- Automated data workflows with Apache Airflow, reducing manual intervention by 50% and ensuring on-time report delivery in banking data processing.
- Enhanced data integration with Snowflake's native support for semi-structured data (like JSON and Parquet), allowing seamless ingestion and processing of diverse data formats in financial reporting.
- Optimized query performance in Snowflake by leveraging clustering keys and materialized views, resulting in faster analytics and reduced compute costs in handling high-volume financial datasets.
- Automated data pipelines within Snowflake using Snowpipe and Snowflake tasks, ensuring real-time data availability and reducing manual intervention in the financial data processing workflow.
- Improved data warehousing and query performance with AWS Redshift and AWS Athena, reducing query times and operational costs through efficient indexing, partitioning, and distribution strategies.

Data Engineer | May 2015 - Dec 2017

Altera Digital Health | Raleigh, NC

- Built and maintained Hadoop-based data lakes supporting 100TB+ of structured and unstructured data.

- Implemented data transformation workflows with AWS Glue, enhancing batch and streaming ETL processes for continuous data availability in the retail domain.
- Designed and optimized Spark jobs on Amazon EMR, improving data processing speed and reducing infrastructure costs for retail data analytics.
- Worked on AWS S3-based data lakes, improving storage efficiency and reducing costs by 20%.
- Integrated data from NoSQL databases like MongoDB into Amazon DynamoDB, offering seamless access to unstructured data for retail analysis.
- Implemented and optimized data security protocols on AWS services, including IAM roles and AWS KMS encryption, to enhance compliance with retail data privacy standards.
- Conducted cost optimization strategies by managing AWS resources effectively, reducing compute and storage costs in the retail domain.

Education

Rowan University, Computer Science - Bachelor's degree

Sep 2011 - Apr 2015